In [1]:
```
pip install numpy scikit-learn
```

```
Requirement already satisfied: numpy in .\anaconda3\Lib\site-packages
(2.4.6)
Requirement already satisfied: scikit-learn in .\anaconda3\Lib\site-packa
ges (1.9.0)
Requirement already satisfied: scipy>=1.10.0 in .\anaconda3\Lib\site-pack
ages (from scikit-learn) (1.18.0)
Requirement already satisfied: joblib>=1.4.0 in .\anaconda3\Lib\site-pack
ages (from scikit-learn) (1.5.3)
Requirement already satisfied: narwhals>=2.0.1 in .\anaconda3\Lib\site-pa
ckages (from scikit-learn) (2.23.0)
Requirement already satisfied: threadpoolctl>=3.5.0 in .\anaconda3\Lib\si
te-packages (from scikit-learn) (3.5.0)
Note: you may need to restart the kernel to use updated packages.
```

In [2]:
```
# Task-04: Hand Gesture or Image Recognition using Machine Learning

# Step 1: Import Required Libraries
from sklearn.datasets import fetch_olivetti_faces
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, classification_report

# Step 2: Load Dataset (Built-in dataset, no manual download required)
print("Loading dataset...")
faces = fetch_olivetti_faces()
X = faces.data  # Image features
y = faces.target  # Labels/Classes

# Step 3: Split Dataset into Training and Testing Sets (80% train, 20% te
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,

# Step 4: Train the Classification Model (SVM Classifier)
print("Training the model, please wait...")
model = SVC(kernel='linear')
model.fit(X_train, y_train)

# Step 5: Evaluate the Model and Calculate Accuracy
y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
print(f"Model Accuracy: {accuracy * 100:.2f}%")
print("\nClassification Report:\n", classification_report(y_test, y_pred,
```

```
Loading dataset...
downloading Olivetti faces from https://ndownloader.figshare.com/files/59
76027 to C:\Users\SHIVAM\scikit_learn_data
Training the model, please wait...
Model Accuracy: 97.50%

Classification Report:
              precision    recall  f1-score   support

           0       0.75      1.00      0.86         3
           1       1.00      1.00      1.00         1
           2       1.00      1.00      1.00         2
           3       1.00      1.00      1.00         4
           4       1.00      1.00      1.00         3
           5       1.00      1.00      1.00         3
           7       1.00      0.83      0.91         6
           8       1.00      1.00      1.00         2
           9       1.00      1.00      1.00         2
          10       1.00      1.00      1.00         2
          11       1.00      1.00      1.00         3
          12       1.00      1.00      1.00         2
          13       1.00      1.00      1.00         1
          14       1.00      1.00      1.00         3
          15       1.00      0.50      0.67         2
          17       0.75      1.00      0.86         3
          18       1.00      1.00      1.00         1
          19       1.00      1.00      1.00         1
          20       1.00      1.00      1.00         1
          21       1.00      1.00      1.00         1
          22       1.00      1.00      1.00         3
          23       1.00      1.00      1.00         2
          24       1.00      1.00      1.00         1
          25       1.00      1.00      1.00         1
          26       1.00      1.00      1.00         4
          27       1.00      1.00      1.00         2
          28       1.00      1.00      1.00         2
          29       1.00      1.00      1.00         1
          32       1.00      1.00      1.00         3
          33       1.00      1.00      1.00         1
          34       1.00      1.00      1.00         1
          35       1.00      1.00      1.00         1
          36       1.00      1.00      1.00         2
          37       1.00      1.00      1.00         2
          38       1.00      1.00      1.00         4
          39       1.00      1.00      1.00         4

    accuracy                           0.97        80
   macro avg       0.99      0.98      0.98        80
weighted avg       0.98      0.97      0.97        80
```

In [ ]: